

BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010

Ethical Analysis Report

Group ID:25-26J-109

Project Title: Multimodal AI Framework for Early Parkinson's Disease Detection

	Identification and Analysis
Data Privacy Concerns	The project utilizes publicly available datasets across multiple modalities (cognitive assessment results, voice recordings, spiral drawing images, and wearable sensor signals) for model training, thereby reducing privacy risks. During the validation phase, biometric data such as cognitive assessment results, voice samples, touchscreen drawing inputs, and wearable sensor readings are collected only after obtaining informed consent from participants in accordance with ethical research guidelines. All collected validation data is anonymized at the point of storage. No personally identifiable information (PII), such as names, national ID numbers, or medical identifiers, is stored together with biometric or behavioral data. The mobile application enforces user-level data isolation through Firebase Authentication with JWT token verification on all backend API calls, and Firestore document-level security rules ensure that each authenticated user can only read and write their own assessment data preventing cross-user data exposure at the database level.
Data Security	All trained model artefacts and assessment data are stored in production-grade secure infrastructure, Google Cloud Firestore for user session data and Google Cloud Run for backend inference services with access controlled via Firebase Authentication and enforced HTTPS/TLS encryption on all client-server communication. Development assets are version-controlled via GitHub with branch-level access controls. Validation data collected from clinic participants is stored exclusively in anonymized form, accessible only to authorized research team members. Backup copies of research data and trained models are maintained to prevent accidental data loss. Any patient validation data collected during testing is securely stored and used exclusively for research purposes. The mobile application and backend system communicate using secure encrypted protocols (HTTPS/TLS) to ensure safe transmission of health-related biometric data.

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	Regular encrypted backups, secure authentication mechanisms, and version control practices are implemented to preserve data integrity and prevent unauthorized modification or loss.
Bias and Fairness	The project acknowledges potential biases in publicly available datasets, including demographic imbalance, linguistic variation, and differences in motor performance across populations. To address this, validation is performed using locally collected data to evaluate generalization across different user groups. The multimodal fusion approach reduces reliance on a single data source, improving robustness by integrating acoustic, linguistic, motor, and cognitive signals. Techniques such as stratified sampling, cross-validation, and class imbalance handling (e.g., SMOTE where applicable) are applied. However, fairness is continuously evaluated rather than assumed to be fully achieved
Transparency and Accountability	The methodologies used for data preprocessing, feature extraction, model training, multimodal fusion, and evaluation are fully documented to ensure transparency and reproducibility. Publicly available datasets, research references, and software tools are properly cited in accordance with academic integrity guidelines. Performance metrics such as accuracy, sensitivity, specificity, F1-score, and AUC are clearly reported to provide transparent evaluation of the framework. The research team remains accountable for the ethical implications of the system and acknowledges that the framework is intended to support early screening and research purposes rather than replace professional medical diagnosis.
Impact on Stakeholders	The project has the potential to benefit individuals at risk of Parkinson's Disease, healthcare professionals, caregivers, and researchers by enabling earlier identification of Parkinson's Disease symptoms through an accessible AI-based screening framework. Early detection may facilitate timely medical consultation and contribute to improved monitoring and management of disease progression. In addition, the wearable and multimodal approach can

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	support remote and home-based healthcare applications, enhancing accessibility to preliminary neurological assessment. Ethical considerations arise regarding potential overreliance on AI-generated predictions or the possibility of incorrect classifications. To mitigate this risk, the system is explicitly designed as a decision-support tool to assist clinical evaluation rather than function as an autonomous diagnostic system. Furthermore, all patient validation data is collected with informed consent and subjected to anonymization procedures. This ensures that risks associated with personal identification, data exposure, or misuse of sensitive health information are significantly minimized.
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